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SCHOOL OF COMPUTER SCIENCE AND APPLICATIONS

Minor Project Report

on

Empowering the Modern Student: An AI-Driven System for Detecting Diverse Stressors and Delivering Actionable Wellness Interventions

A Project Report submitted in partial fulfilment of the requirements for the award of the Degree of Master of Computer Applications - MCA

Submitted by

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CERTIFICATE

This is to certify that the Minor project work entitled "**Empowering the Modern Student: An AI-Driven System for Detecting Diverse Stressors and Delivering Actionable Wellness Interventions**" submitted to the School of Computer Science and Applications, REVA University in partial fulfilment of the requirements for the award of the Degree of **Master of Computer Applications** in the academic year 2025-2026 is a record of the original work done by **ALSAA FATIMA (R24DE019) and APARNA.C (R24DE023)** under my supervision and guidance. The project report has been approved as it satisfies the academic requirements in respect of Semester III Project work prescribed for the said Degree and this Minor project work has not formed the basis for the award of any Degree / Diploma / Associate ship / Fellowship or similar title to any candidate of any University.

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Signature with date 24/2/26

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DECLARATION

We, ALSAA FATIMA (R24DE019) and APARNA.C (R24DE023) third semester students of Master of Computer Applications belonging to School of Computer Science and Applications, REVA University, declare that this Project work entitled **"Empowering the Modern Student: An AI-Driven System for Detecting Diverse Stressors and Delivering Actionable Wellness Interventions"** is the result of the Project work done by us under the supervision of Dr. P. Sree Lakshmi, Assistant Professor, School of Computer Science and Applications.

We are submitting this Project work in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications by REVA University, Bangalore during the academic year 2025-26.

We further declare that this Project report or any part of it has not been submitted for the award of any other Degree / Diploma of this University or any other University / Institution.

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Certified that this project work submitted by Alsaa Fatima and Aparna . C has been carried out under my guidance and the declaration made by the candidates is true to the best of my knowledge.

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ABSTRACT

Mental stress has become a significant issue among students and working professionals due to increasing academic workload, career pressure, and social expectations. Continuous exposure to such stress can negatively affect mental health, productivity, and overall well-being. Traditional methods used for stress detection mainly depend on questionnaires or single-source data analysis, which often produce subjective results and are not suitable for accurate or real-time monitoring. Therefore, there is a need for an intelligent system that can detect stress levels more effectively using advanced technologies.

The main objective of this project is to develop an AI-driven multimodal system for detecting and analyzing mental stress levels. The proposed system uses multiple data sources such as speech signals, textual information, and physiological signals to identify stress patterns. Machine learning and deep learning techniques are applied to process and analyze the collected data. Natural Language Processing techniques are used to analyze textual data and identify stress-related expressions. Speech emotion recognition techniques are used to analyze voice signals and detect emotional variations. Physiological signals are also considered as objective indicators for measuring stress levels.

The system integrates all the collected information using a multimodal fusion approach, which improves the overall accuracy and reliability of stress detection. The fusion-based approach helps in overcoming the limitations of unimodal systems and provides more consistent results. The results obtained from the proposed system demonstrate that the multimodal approach is more effective in detecting stress compared to traditional single-mode methods.

The developed system can be used as a supportive tool for monitoring mental stress among students and professionals. It can help in early detection of stress and provide insights for improving mental well-being. The project demonstrates the usefulness of artificial intelligence techniques in developing intelligent healthcare and wellness monitoring solutions.

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CHAPTER - I

INTRODUCTION

1.1 Introduction to the Project

Mental stress has become a common problem among students and working professionals due to increasing academic pressure, workload, and social expectations. Continuous stress can negatively affect mental health, productivity, and overall well-being. Early detection of stress is important in order to prevent serious health problems and improve quality of life.

Traditional stress detection methods mainly depend on questionnaires, interviews, and clinical assessments. Although these methods are useful, they are time-consuming and often subjective. They also cannot provide real-time stress monitoring. With the advancement of Artificial Intelligence and Machine Learning, it is possible to develop automated systems that can detect stress more accurately and efficiently.

This project proposes an AI-driven multimodal system for detecting mental stress levels using speech signals, text inputs, and physiological signals. The system analyzes user input through machine learning and deep learning models to identify stress patterns and classify stress levels into Low, Moderate, and High categories. The multimodal approach improves the reliability and accuracy of stress detection compared to traditional unimodal systems.

The proposed system provides a user-friendly interface that allows users to submit voice recordings and textual descriptions of their emotional state. The system processes the inputs and displays the detected stress level along with analysis results. This project aims to provide an intelligent and efficient solution for mental stress detection and monitoring.

1.2 Statement of the Problem

Mental stress has increased significantly due to academic pressure, professional workload, and lifestyle changes. Existing stress detection systems often rely on single sources such as speech patterns, textual data, or physiological signals, which may produce inaccurate results due to noise, environmental variations, and individual differences.

Therefore, there is a need for an intelligent system that can accurately detect stress levels using multiple input sources. A multimodal system that combines speech, text, and physiological signals can improve the accuracy and reliability of stress detection and provide better results compared to traditional methods.

1.3 System Specifications

➢ Hardware Requirements

• **Device**: Desktop / Laptop (Intel i3 / Intel i5 Processor @ 3.0 GHz or higher, 8–16 GB RAM)

• **Storage:** Minimum 100 GB HDD/SSD for project files and datasets

• **Graphics**: Integrated GPU (suitable for small Machine Learning tasks)

• **Microphone:** Required for speech-based stress detection

➢ Software Requirements

• **Operating System:** Windows 10 / Windows 11

• **Programming Language:** Python 3.x, JavaScript

• **Frontend Technologies:** HTML, CSS, ReactJS, Tailwind CSS

• **Backend Technologies:** Python (Machine Learning Model)

• **IDE / Development Environment:** Visual Studio Code / Jupyter Notebook

➢ **Libraries / Frameworks:**

- pandas
- NumPy
- matplotlib
- scikit-learn
- Librosa
- ReactJS
- Node.js

➢ **Database :**

- MongoDB and Postman

➢ **Dataset Requirements**

• Dataset Type: Stress Detection Dataset

• Format: CSV files and Audio files

• Purpose: Used for training and testing the machine learning model for stress prediction.

• Storage Requirement: Minimum 1–2 GB dataset storage.

➢ **Network Requirements**

• Internet connection required for:

- Downloading datasets
- Installing libraries
- Running npm packages

CHAPTER - II

LITERATURE SURVEY

2.1 Introduction

This chapter provides a comprehensive review of research on mental stress detection using machine learning and deep learning techniques. The primary objectives of this review are to:

- Examine the methods and approaches used in stress classification,
- Identify limitations of current systems,
- Highlight gaps in existing research, and
- Assess the potential benefits of multimodal stress detection frameworks.

Thirty research studies have been analyzed, covering speech-based stress detection, physiological signal monitoring, wearable devices, emotion recognition, and multimodal data fusion. The review discusses methodologies, datasets, feature extraction techniques, classifier performance, and limitations of each study.

2.2 Domain Overview - AI-Based Mental Stress Detection

Artificial intelligence approaches enable automated stress recognition by analyzing behavioural, vocal, and physiological patterns. Traditional stress assessment methods, such as ECG, cortisol measurement, questionnaires, and clinical evaluation, are reliable but often:

- Are costly,
- Cannot provide real-time feedback,
- Are intrusive, and
- Do not support continuous monitoring.

Machine learning and deep learning offer scalable solutions for automated, continuous, and multimodal stress evaluation. Speech-based systems extract acoustic features like MFCC, pitch, jitter, and energy, whereas physiological systems rely on HRV, ECG, EEG, and galvanic skin response. Multimodal fusion, integrating multiple input types, has demonstrated improved classification accuracy and robustness compared to single-modality approaches.

2.3 Key Techniques and Technologies

2.3.1 Machine Learning Algorithms

These include, but are not limited to, SVM, Random Forest, KNN, Logistic Regression, and ANN, all of which have been widely used for stress detection. These are the methods that perform well in structured datasets and usually serve as baseline models when comparing with deep learning approaches [1]–[6].

2.3.2 Deep Learning Approaches

Deep architectures like CNN, LSTM, GRU, and hybrid CNN-LSTM or CNN-RNN automatically extract complex patterns from speech and physiological signals. All of these methods generally result in higher accuracy in classification ,but require a bigger dataset and much more computational resources [7]–[12].

2.3.3 Feature Extraction Techniques

Feature extraction is thus an important step in stress classification. Some of the more common features selected for use in this regard are MFCC, jitter, pitch, entropy, spectral energy, HRV, and galvanic responses. Proper feature selection leads to higher performance by a classifier with reduced computational complexity as well [1]–[8].

2.3.4 Multimodal Fusion

Multimodal techniques are based on various inputs: speech, facial expressions, EEG, ECG, HRV, and wearable sensors. Different fusion strategies, such as early, late, and hybrid feature integration, provide greater reliability and reduce some disadvantages of unimodal systems [5]–[8].

2.4 Review of 30 Related Research Works

The following summarizes key findings from thirty studies on stress detection:

- Multimodal stress detection using wearable sensors and CNN achieved high accuracy but was limited by noise sensitivity [1].
- Adaptive AI-based cognitive therapy systems combining facial and voice analysis reached 92–93% accuracy, but data bias and environmental factors affected performance [2].
- Multi-stage stress detection using speech and lexical features improved F1-score but was constrained by processing speed [3].

- Real-time EEG-based stress detection using hybrid LSTM achieved higher accuracy, though dataset diversity was limited [4].
- CNN-based facial and vocal stress detection obtained 96% accuracy, yet cross-modal synchronization posed challenges [5].
- Workplace multimodal analysis (audio, facial, text) enhanced reliability but was affected by lighting and user dependency [6].
- TEO + MFCC speech feature CNN classifiers improved classification over standard MFCC but required more diverse datasets [7].
- Voice-based machine learning stress analysis demonstrated feasibility but was limited by single-language datasets [8].
- Combined speech and physiological data frameworks improved stress detection accuracy but required large-scale validation [9].
- Deep learning techniques on wearable data provided continuous monitoring; however, computational costs were high [10].
- Speech breathing patterns modeled via bi-LSTM and XGBoost showed potential but required more physiological inputs [11].
- CNN-LSTM hybrid models for emotion detection achieved 71% accuracy and highlighted high-arousal emotion detection strength [12].
- Text-based stress analysis using Word2Vec and Random Forest reached high accuracy for multiple languages but lacked multimodal integration [13].
- Chatbot-assisted stress management frameworks were effective but limited by dataset diversity [14].
- Text-based TF-IDF vectorization with Random Forest achieved 85% accuracy but required semantic improvements [15].
- Autonomic-signal-based stress detection with kernel PCA and k-NN achieved reliable classification but required multiple devices [16].
- Multimodal stress evaluation using DASS-42 questionnaires and facial emotion recognition improved deception resistance but had moderate accuracy [17].
- Hybrid CNN, LSTM, and transformer models for adolescent stress detection provided early prediction potential [18].
- Deep regression CNNs integrating facial and physiological data captured subtle daily stress with F1-score improvement [19].
- Multimodal fusion with ANN, Random Forest, GBM, and XGBoost yielded higher accuracy but faced data heterogeneity issues [20].

- HRV-based multi-class stress recognition using LSTM demonstrated robustness but lacked real-time personalization [21].
- Graph Neural Networks with temporal convolutions enhanced multimodal stress detection but required dataset adaptation [22].
- Federated learning-based multimodal workplace stress detection ensured privacy but had moderate classification accuracy [23].
- Facial CNNs achieved over 90% accuracy for real-time stress monitoring, highlighting non-invasive monitoring feasibility [24].
- Feedforward neural networks integrating accelerometer, GSR, and HRV delivered 90% accuracy in real-time settings [25].
- Speech acoustic-based stress recognition demonstrated effective classification but suffered from speaker variability [26].
- CNN-LSTM voice emotion recognition frameworks improved reliability but required balanced datasets [27].
- Random Forest with HRV classification achieved robust results but was susceptible to overfitting [28].
- SVM-based speech stress recognition validated the relationship between vocal changes and stress [29].
- Statistical behavioral modeling provided foundational insights into early algorithmic stress detection [30].

Table 1 : Comparative Analysis of various techniques for Stress Detection of Students from Literature

Paper	**Technique**	**Dataset/Source**	**Result / Accuracy**	**Limitation**
1	Multimodal CNN	Wearables	High	Noise sensitivity
2	AI Cognitive Therapy	Speech + Facial	92–93%	Data bias, environment
3	Speech + Lexical CNN	SAVEE, CREMA-D	Good	Processing speed
4	EEG + LSTM	EEG	High	Limited dataset
5	CNN + SVM	FER-2013,	96%	Cross-modal sync

		RAVDESS		
6	Bi-LSTM + ViT	Audio + Facial + Text	87–80%	Lighting, user-dependency
7	TEO + MFCC CNN	RAVDESS	95%	Dataset diversity
8	Voice ML	Audio	Good	Single-language dataset
9	Multimodal ML	Speech + Physiological	High	Large-scale validation
10	Deep Learning	Wearables	Continuous	High computation
11	Bi-LSTM + XGBoost	Speech Breathing	64%	Limited physiological features
12	CNN-LSTM	RAVDESS	71%	Low-arousal emotion accuracy
13	Random Forest + Word2Vec	Text	99%	No multimodal integration
14	Chatbot + Facial CNN	FER-2013	85%	Dataset diversity
15	TF-IDF + RF	Text	85%	Semantic improvement needed
16	Kernel PCA + k-NN	Wearables	F1 0.66	Multiple devices
17	Questionnaire + CNN	FER-2013	Moderate	Dataset constraints
18	CNN + LSTM + Transformer	Wearable + Questionnaire	High	Needs privacy validation
19	DR-CNN	Facial + Physiological	F1 80–96%	Context-specific dataset
20	ANN + RF + GBM	ECG + Audio	94–97%	Data heterogeneity
21	LSTM	HRV	High	Limited real-time personalization
22	GNN + T-GCN	ECG + Behavioral	92%	Dataset adaptation
23	Federated Learning	Multi-modal	Good	Moderate accuracy

		Workplace		
24	CNN Facial	AffectNet + FER	>90%	Population generalization
25	FNN	ECG + GSR	90%	Dataset bias
26	Acoustic ML	Speech	Moderate	Speaker variability
27	CNN-LSTM	Speech + Face	Good	Dataset imbalance
28	Random Forest	HRV	Good	Overfitting
29	SVM	Speech	Moderate	Small dataset
30	Statistical Model	Behavioral	Baseline	Low accuracy

2.6 Gaps Identified in Literature

- Predominant use of single-modality systems.
- Noise susceptibility in speech-based models.
- Limited generalization across the different users.
- The lack of real-time multimodal frameworks.
- High cost and power consumption by wearable sensors.
- Purely speech-based solutions have difficulties in noisy conditions.
- Limited availability of large multimodal datasets.
- Manual feature extraction dependence in many studies.
- Accuracy in deep learning strongly depends on data volume.

2.7 Motivation for Proposed Work

The gaps identified in previous studies highlight the need for a multimodal AI-based stress detection framework that:

- Integrates multiple stress indicators (speech, physiological signals, wearable data),
- Operates in real-time,
- Minimizes reliance on manual feature engineering,

- Improves user generalization, and
- Enhances overall classification performance.
- This motivates the development of the proposed speech + physiological multimodal stress detection system.

2.8 Chapter Summary

This chapter reviewed thirty research studies on the detection of stress, speech, physiological signals, wearable devices, and multimodal fusion approaches. Although remarkable progresses have been achieved in terms of accuracy and model architecture, limitations remain regarding generalization, real-time implementation, and multimodal integration. The findings above motivate developing a robust, real-time multi-modal stress detection system that could find practical applications in mental health.

CHAPTER - III

SYSTEM ANALYSIS

3.1 Existing System

In the existing systems, mental stress detection is mainly performed using traditional methods such as questionnaires, interviews, and clinical evaluations. These methods depend on the user's responses and therefore may not always provide accurate results.

Some existing stress detection systems use single types of data such as speech signals, text analysis, or physiological signals. These unimodal systems detect stress using only one input source, which reduces reliability and accuracy.

Most existing systems are not capable of real-time monitoring and require manual analysis. Many systems also depend on expensive medical devices and specialized equipment, making them difficult to use in everyday situations.

Therefore, existing stress detection methods are not fully efficient for continuous and automatic stress monitoring.

3.2 Limitations of the Existing System

The existing stress detection systems have several limitations:

- Depend heavily on questionnaires and manual evaluation
- Results may be subjective and inaccurate
- Mostly use single data source (speech or text or physiological signals)
- Cannot provide real-time stress detection
- Require expensive medical equipment
- Not suitable for continuous monitoring
- Less accuracy compared to multimodal systems
- Limited accessibility for normal users

3.3 Proposed System

The proposed system is an **AI-driven multimodal stress detection system** designed to identify stress levels using multiple types of data such as speech, text, and physiological signals.

The system analyzes user inputs and detects stress levels using artificial intelligence techniques. The system integrates multiple inputs to improve detection accuracy and reliability.

The system stores user information and stress data using a database system such as **MongoDB**, and APIs are tested using **Postman**. The proposed system aims to provide automatic and efficient stress detection.

The system also provides wellness suggestions based on detected stress levels to help users improve their mental well-being.

3.4 Advantages of the Proposed System

The proposed system provides several advantages:

- Automatic stress detection
- Improved accuracy using multiple data sources
- Real-time stress monitoring
- User-friendly system
- Efficient data storage using MongoDB
- Reliable API testing using Postman
- Reduced manual effort
- Better performance compared to existing systems
- Helps users manage stress effectively

CHAPTER - IV

SYSTEM DESIGN

4.1 High Level Design

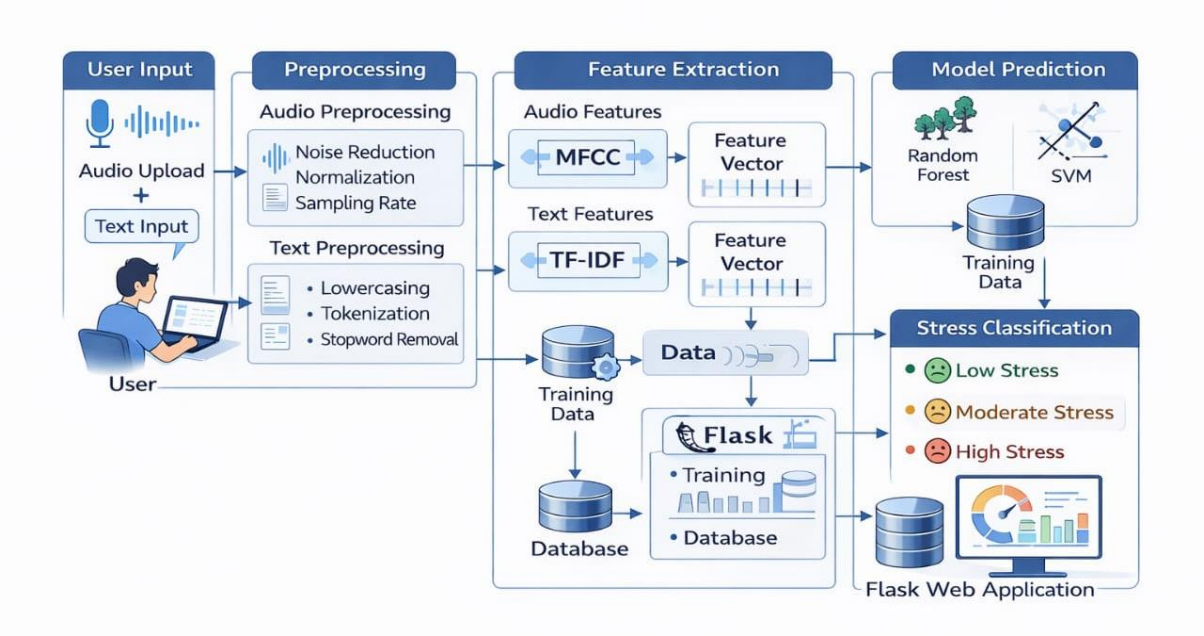

Fig 4.1 Architectural Diagram

The architectural design represents the overall structure of the proposed AI-driven stress detection system. The system accepts user input in the form of **audio and text data** and processes it to detect stress levels.

Step 1: User Input

The user provides input to the system in the form of:

- Audio upload (speech input)
- Text input

These inputs are used to detect stress levels.

Step 2: Preprocessing

The input data is cleaned and prepared before analysis.

Audio Preprocessing includes:

- Noise reduction

- Normalization
- Sampling rate adjustment

Text Preprocessing includes:

- Lowercasing
- Tokenization
- Stopword removal

Step 3: Feature Extraction

Important features are extracted from the input data.

Audio Features:

- MFCC (Mel Frequency Cepstral Coefficients)

Text Features:

- TF-IDF (Term Frequency – Inverse Document Frequency)

These features are converted into feature vectors.

Step 4: Model Prediction

Machine learning models are used to predict stress levels.

The models used include:

- Random Forest
- Support Vector Machine (SVM)

Training data is used to train the models.

Step 5: Stress Classification

The system classifies stress levels into:

- Low Stress
- Moderate Stress
- High Stress

Step 6: Database and Web Application

The system stores data and results in the database. The system is implemented as a web application and APIs are tested using Postman.

4.2 Low Level Design

The low-level design describes the internal working of each module in the stress detection system. It explains how the input data is processed step by step to detect stress levels.

I. User Input Module

The user provides input to the system in the form of:

- Audio input (speech data)
- Text input

The system receives the input and sends it to the preprocessing module.

II. Preprocessing Module

The preprocessing module cleans and prepares the input data before feature extraction.

Audio preprocessing includes:

- Noise reduction
- Normalization
- Sampling rate adjustment

Text preprocessing includes:

- Lowercase conversion
- Tokenization
- Stopword removal

After preprocessing, the data is sent to the feature extraction module.

III. Feature Extraction Module

This module extracts important features from the input data.

Audio features:

- MFCC features are extracted from speech signals.

Text features:

- TF-IDF features are extracted from text input.

The extracted features are converted into feature vectors for model training and prediction.

IV. Model Module

The machine learning models analyze the feature vectors and predict stress levels.

The models used include:

- Random Forest
- Support Vector Machine (SVM)

The models are trained using training datasets and used for stress prediction.

V. Database Module

The database module stores:

- User input data
- Training data
- Stress prediction results

MongoDB is used for storing the data.

VI. Result Module

The final output is displayed to the user through the web application.

The system classifies stress into:

- Low Stress
- Moderate Stress
- High Stress

The results are stored in the database and shown on the user interface.

CHAPTER -V

DATA COLLECTION AND PREPARATION

5.1 Data Sources

The data used in this project was collected from publicly available benchmark datasets and sample user inputs. The dataset includes both audio data and text data for stress level prediction. These datasets are widely used in research and provide labeled emotional data suitable for stress-level mapping.

5.1.1 RAVDESS (Ryerson Audio-Visual Database of Emotional Speech)

Dataset Name: RAVDESS
Source: Ryerson University
Official Link: https://zenodo.org/record/1188976

RAVDESS contains 1440 high-quality speech recordings in .wav format. The dataset includes eight emotional categories: neutral, calm, happy, sad, angry, fearful, disgust, and surprised. It is widely used for speech emotion recognition and is suitable for speech-based stress detection.

5.1.2 TESS (Toronto Emotional Speech Set)

Dataset Name: TESS
Source: University of Toronto
Official Link: https://tspace.library.utoronto.ca/handle/1807/24487

The TESS dataset consists of 2800 speech recordings spoken by female speakers and labeled with seven emotional categories. This dataset improves the robustness and generalization capability of the stress detection model.

5.1.3 Emo-DB (Berlin Emotional Speech Database)

Dataset Name: Emo-DB
Source: Technical University of Berlin
Official Link: http://emodb.bilderbar.info/start.html

Emo-DB contains 535 emotional speech recordings in the German language, categorized into seven emotional classes. This dataset is useful for evaluating cross-language performance of the stress detection system.

5.1.4 Sample User Input Data

In addition to benchmark datasets, sample user input data in the form of text and audio inputs was collected to test real-time stress prediction functionality.

5.2 Dataset Composition

The overall dataset used in this project includes:

- Audio recordings for speech-based stress analysis
- Text data representing stress-related expressions
- Training and testing datasets for machine learning models
- Sample user input data

Total Audio Samples Used:

- RAVDESS: 1440
- TESS: 2800
- Emo-DB: 535

Total = 4,775 audio recordings

All audio files are in .wav format and were recorded under controlled conditions to ensure clarity and consistency.

5.3 Data Profiling

Data profiling was performed to analyze the structure, size, and quality of the collected datasets before preprocessing and model training.

5.3.1 Audio Data Profiling

The following aspects were examined during audio data profiling:

- Total number of audio samples: 4,775
- File format consistency (.wav)
- Sampling rate uniformity
- Audio duration consistency
- Presence of corrupted or incomplete files
- Duplicate file detection
- Verification of emotion labels

The dataset contains emotional speech samples labeled as neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

5.3.2 Emotion-to-Stress Mapping

Since the datasets provide emotion labels rather than direct stress labels, an emotion-to-stress mapping strategy was applied:

Emotion Category	Stress Level
Neutral / Calm	Low Stress
Sad / Fear	Moderate Stress
Angry / Disgust	High Stress

This mapping enabled the effective use of emotion datasets for stress level prediction.

5.3.3 Text Data Profiling

Text data was analyzed to ensure:

- No missing or null values
- Proper text encoding (UTF-8)
- Removal of duplicate entries
- Consistent labeling of stress categories

Data profiling helped in improving dataset quality and ensuring balanced representation of stress levels.

5.4 Data Cleaning and Preprocessing

After profiling, data cleaning and preprocessing were performed to enhance dataset quality before feature extraction and model training.

5.4.1 Audio Data Preprocessing

The following preprocessing steps were applied to audio data:

- Noise reduction to remove background disturbances
- Signal normalization to standardize amplitude levels
- Sampling rate adjustment for uniformity

- Silence trimming to remove unnecessary pauses

These steps improved the quality and consistency of speech signals.

5.4.2 Text Data Preprocessing

The following preprocessing steps were applied to text data:

- Conversion of text to lowercase
- Tokenization
- Removal of stopwords
- Removal of special characters and punctuation
- Lemmatization or stemming (where applicable)

These steps prepared the text data for feature extraction.

5.5 Data Splitting and Storage

After preprocessing, the dataset was divided into:

- 70% Training data
- 20% Testing data
- 10% Validation data

The processed data and user inputs were stored and managed using a MongoDB database, ensuring efficient access, scalability, and data integrity.

5.6 Summary

This chapter described the data sources, profiling, cleaning, and preprocessing steps used in the project. Publicly available benchmark datasets such as RAVDESS, TESS, and Emo-DB were combined with text-based stress data and sample user inputs to create a reliable dataset for training and evaluating the proposed stress detection system.

CHAPTER -VI

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the characteristics of the dataset before building the stress detection model. EDA helps in identifying patterns, trends, and relationships in the data.

In this project, exploratory analysis was performed on both **audio data and text data** to understand the distribution of stress-related features.

EDA helps in:

- Understanding dataset structure
- Identifying patterns in data
- Detecting missing values
- Understanding feature importance
- Improving model performance

6.1 Data Visualization Techniques

Data visualization techniques are used to represent data in graphical form for better understanding.

The following visualization techniques were used:

- Bar charts to represent stress level distribution
- Pie charts to show percentage of stress categories
- Histograms to understand data distribution
- Line graphs to observe trends in data

These visualizations help in understanding the relationship between input data and stress levels.

Data visualization makes it easier to interpret large amounts of data and identify important patterns.

6.2 Univariate and Bivariate Analysis

Univariate Analysis

Univariate analysis is used to analyze a single variable at a time.

In this project, univariate analysis was performed on:

- Stress level categories
- Audio features
- Text features

This analysis helped in understanding the distribution of individual variables.

Bivariate Analysis

Bivariate analysis is used to study the relationship between two variables.

In this project, bivariate analysis was performed to understand:

- Relationship between audio features and stress levels
- Relationship between text features and stress levels

This analysis helped in identifying important features for stress detection.

CHAPTER - VII

METHODOLOGY

7.1 Data Models

The proposed system follows a **multimodal machine learning approach** for stress detection. The system integrates both **audio-based speech signals** and **text-based user input** to improve prediction accuracy. The overall methodology is designed in multiple stages: input acquisition, preprocessing, feature extraction, model training, and stress classification.

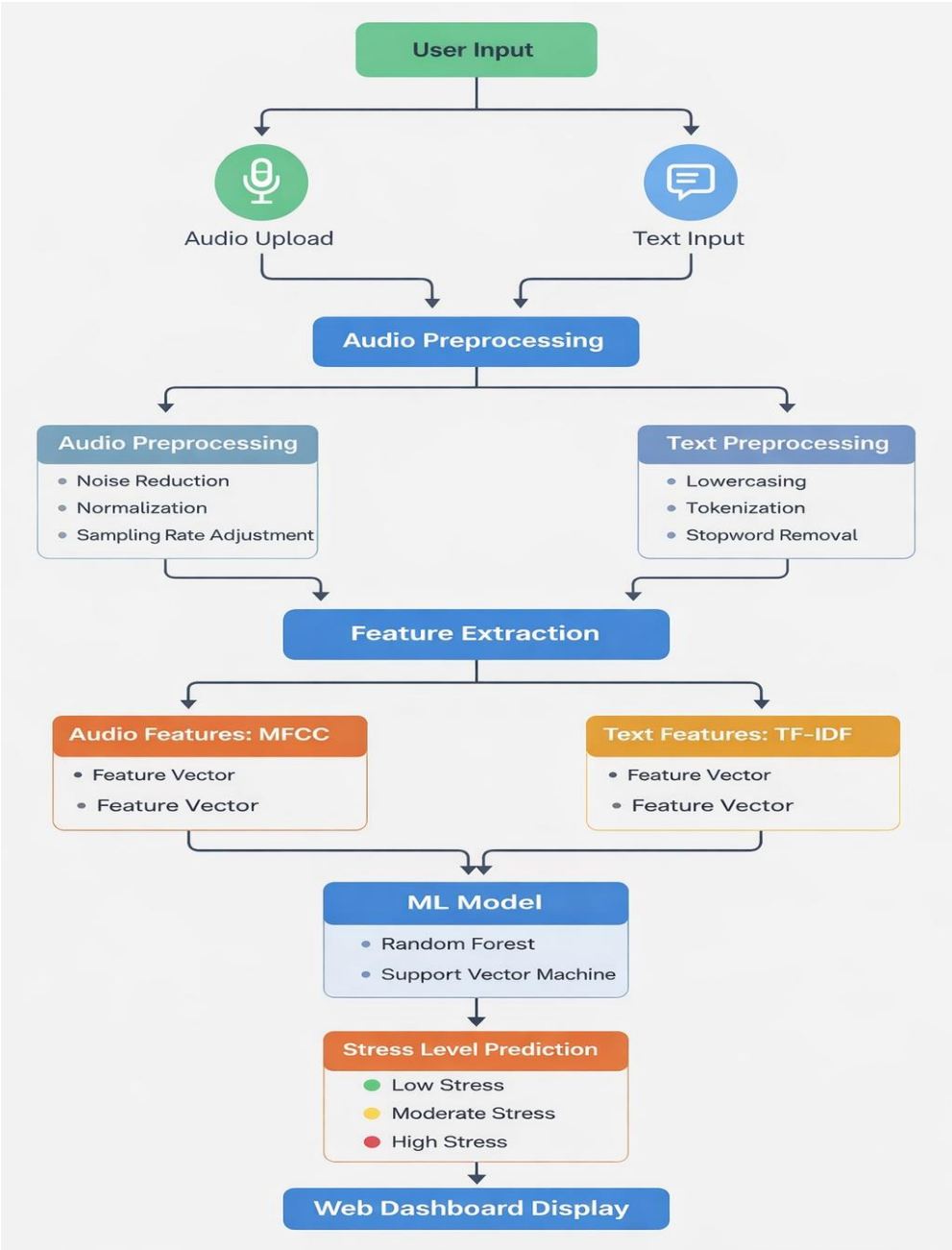

Fig 7.1 Block diagram of the proposed stress level prediction system.

7.1.1 Multimodal Input Model

The system accepts two forms of input:

1. **Audio Input (Voice Recording)**
2. **Text Input (Typed Message)**

The user may provide either one or both inputs. The multimodal nature of the system allows better contextual understanding of stress levels.

7.1.2 Audio Processing Model

The audio data undergoes multiple preprocessing steps before feature extraction.

a) Noise Reduction

Background noise present in recordings is removed using signal filtering techniques. This improves speech clarity and model accuracy.

b) Normalization

Audio amplitude levels are standardized to ensure uniform loudness across samples.

c) Sampling Rate Adjustment

All audio files are converted to a consistent sampling rate to maintain uniform feature representation.

After preprocessing, **MFCC (Mel-Frequency Cepstral Coefficients)** are extracted.

MFCC Feature Extraction

MFCC captures:

- Pitch variations
- Speech tone
- Frequency distribution
- Emotional cues

The extracted MFCC coefficients are converted into a **numerical feature vector** representing speech characteristics related to stress.

7.1.3 Text Processing Model

Text input is processed using Natural Language Processing (NLP) techniques.

a) Lowercasing

All text is converted to lowercase to ensure uniformity.

b) Tokenization

The text is split into individual words or tokens.

c) Stopword Removal

Common words such as "is", "the", "and" are removed as they do not contribute significantly to emotional meaning.

After preprocessing, **TF-IDF (Term Frequency – Inverse Document Frequency)** is applied.

TF-IDF Feature Extraction

TF-IDF assigns weights to words based on:

- Word frequency in a document
- Word rarity across all documents

The result is a **weighted numerical vector** representing emotional intensity and contextual importance.

7.1.4 User and Admin Model

The proposed stress detection system consists of two main modules:

1. **User Module**
2. **Admin Module**

These modules ensure smooth system operation and proper data management.

User Module

The user module allows individuals to interact with the stress detection system through the web interface.

The main functionalities include:

• User Registration and Login
Users can create an account and log in securely using their credentials.

• Stress Input Submission
Users can provide:

- Voice recordings for speech-based stress detection
- Text messages for text-based stress detection

• Stress Prediction View
After processing, the system displays the predicted stress level:

- Low Stress
- Moderate Stress
- High Stress

• Result History
Users can view their previous stress prediction results stored in the database.

The user module provides an easy and interactive interface for stress analysis.

Admin Module

The admin module is responsible for managing the system and monitoring user activities.

The main functionalities include:

• Admin Login
The administrator logs in using secure credentials.

• User Management
Admin can:

- View registered users
- Remove inactive users if required

• Dataset Management
Admin can:

- Upload training datasets
- Update datasets when required

• Model Monitoring
Admin can monitor model performance and system functionality.

• Result Monitoring
Admin can view stress prediction records generated by users.

The admin module ensures proper control and maintenance of the stress detection system.

7.2 Model Selection

The system employs classical supervised machine learning algorithms:

1. Random Forest

Random Forest is an ensemble learning method that:

- Uses multiple decision trees
- Reduces overfitting
- Improves prediction stability

It performs well with high-dimensional data such as MFCC and TF-IDF feature vectors.

2. Support Vector Machine (SVM)

SVM is effective in:

- High-dimensional feature spaces
- Binary and multi-class classification
- Handling non-linear decision boundaries

SVM constructs an optimal hyperplane that separates stress categories.

Reason for Selecting These Models

- Efficient for structured feature vectors
- Computationally less expensive than deep learning
- Suitable for moderate-sized datasets
- Provide high accuracy in classification tasks

7.3 Model Building

The model building process consists of the following steps:

Step 1: Feature Vector Preparation

MFCC and TF-IDF outputs are converted into numerical arrays.

Step 2: Dataset Splitting

The dataset is divided into:

- Training Set (80%)
- Testing Set (20%)

Step 3: Model Training

The training dataset is fed into:

- Random Forest classifier
- Support Vector Machine classifier

The model learns patterns associated with:

- Low stress
- Moderate stress
- High stress

Step 4: Model Evaluation

Performance metrics used:

- Accuracy
- Precision
- Recall
- F1-Score

The best-performing model is selected for deployment.

7.4 Results

Final Output Results

This chapter presents the final output obtained from the developed system titled **"Empowering the Modern Student: An AI-Driven System for Detecting Diverse Stressors and Delivering Actionable Wellness Interventions."** The system analyzes user input and classifies student stress levels using trained machine learning models. Only the final result outputs are presented in this chapter.

Stress Level Classification Output

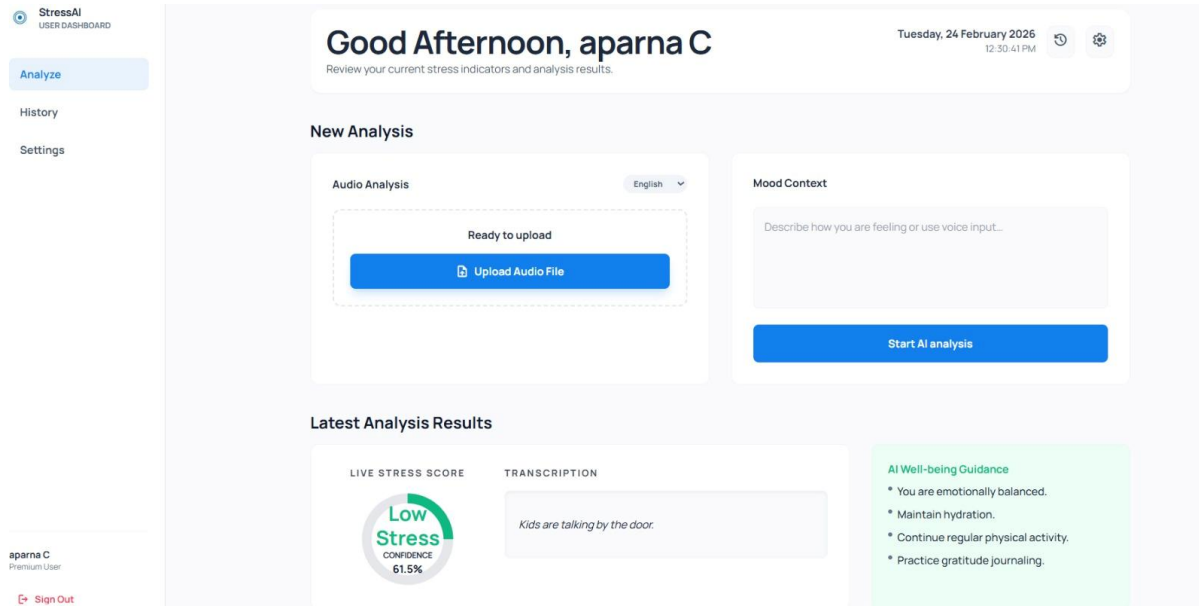

Figure 7.4.1: Stress Detection Result – Low Stress with audio

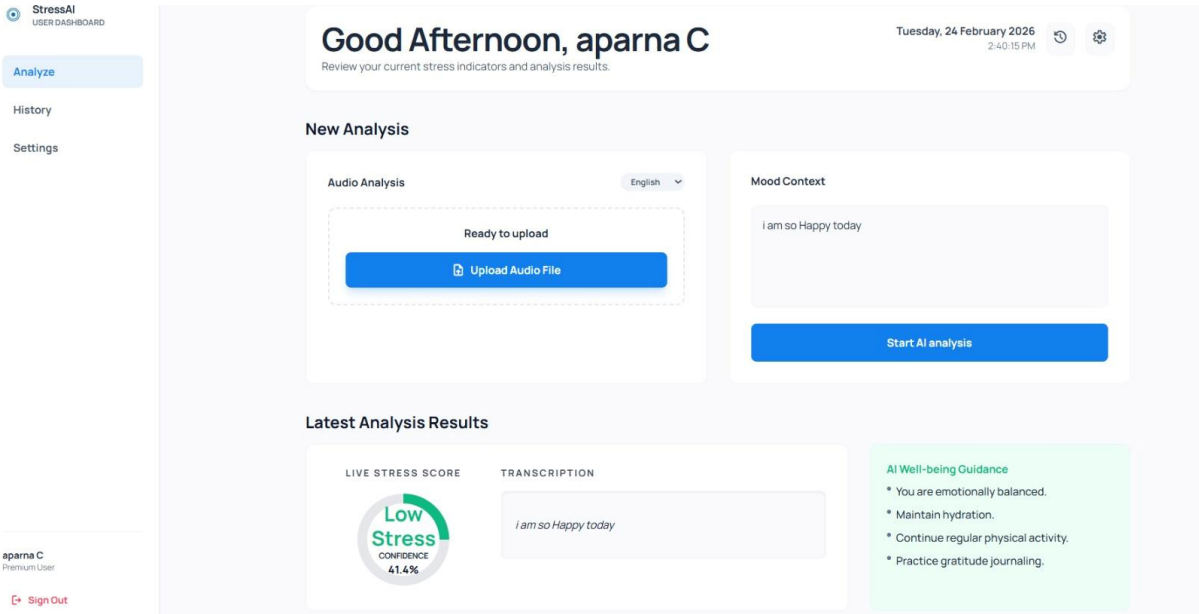

Figure 7.4.2: Stress Detection Result – Low Stress with text

The system successfully analyzes the input data and classifies the student's stress level as **Low Stress**. The result is displayed through a user-friendly web application interface.

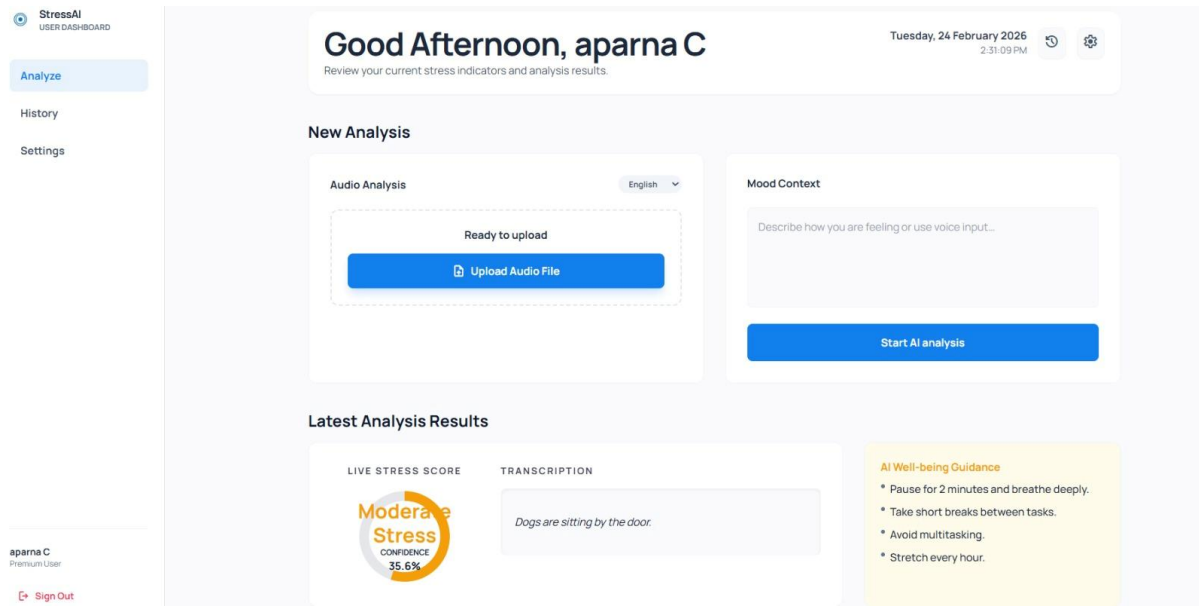

Figure 7.4.3: Stress Detection Result – Moderate Stress with audio

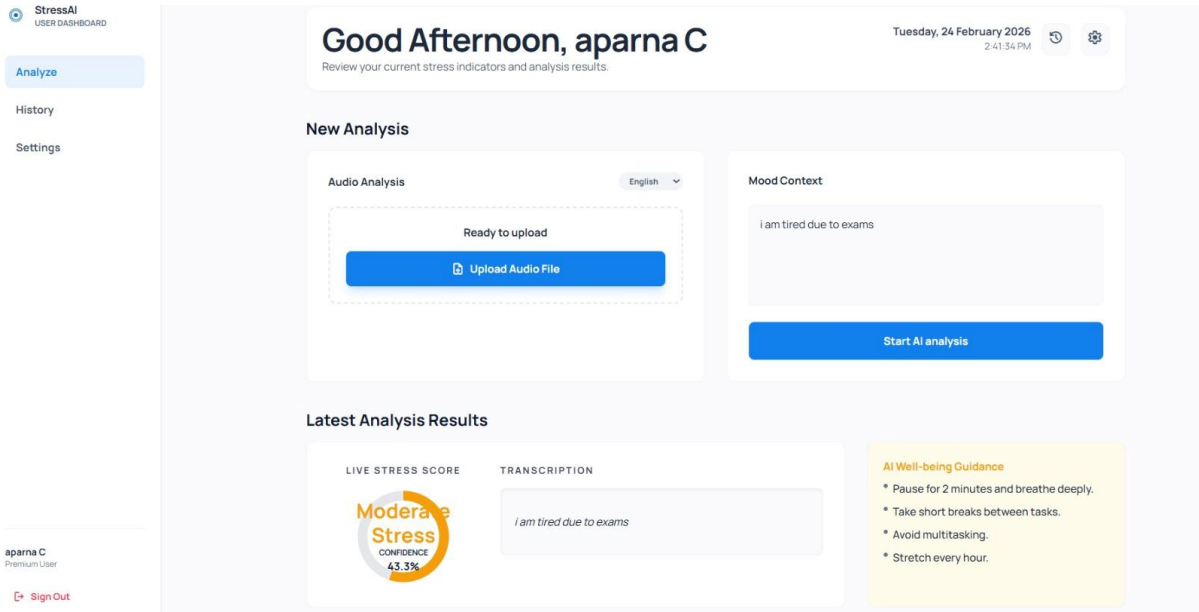

Figure 7.4.4: Stress Detection Result – Moderate Stress with text

For moderate stress conditions, the system accurately predicts the stress level as **Moderate Stress**, indicating effective stressor detection and analysis.

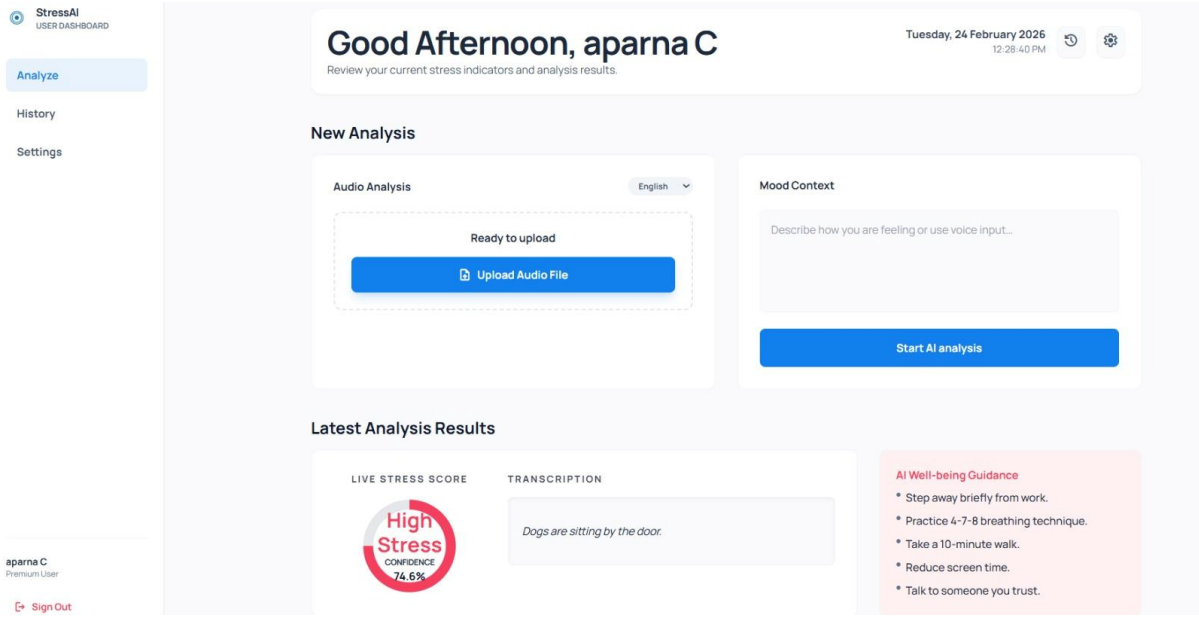

Figure 7.4.5: Stress Detection Result – High Stress with audio

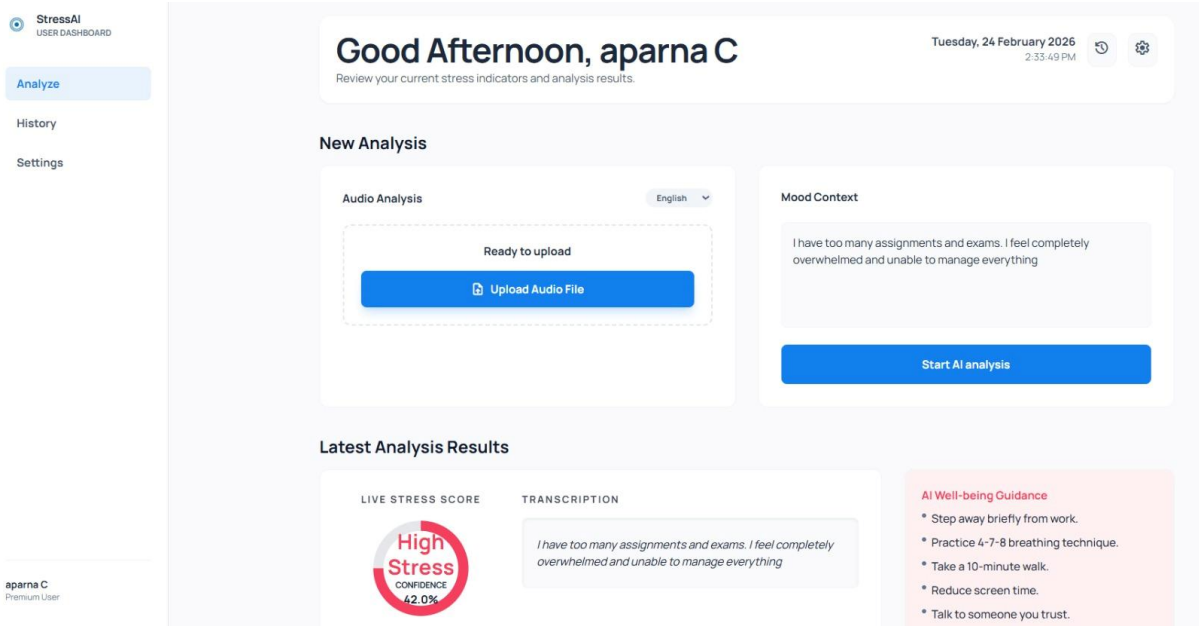

Figure 7.4.6: Stress Detection Result – High Stress with text

The system identifies high stress conditions and classifies the output as **High Stress**, enabling timely awareness and support for the student.

Result Summary

The final results demonstrate that **"Empowering the Modern Student: An AI-Driven System for Detecting Diverse Stressors and Delivering Actionable Wellness Interventions"** is capable of:

- Detecting diverse student stress levels accurately
- Classifying stress into **Low, Moderate, and High** categories
- Providing clear and interpretable outputs
- Supporting student wellness through timely stress identification

7.5 Observations:

- Multimodal input improves classification reliability.
- Audio features capture emotional tone effectively.
- Text features capture contextual emotional content.
- Random Forest provided stable performance across datasets.
- SVM showed strong boundary separation for moderate stress cases.

The integration of both modalities increases the robustness of the system compared to unimodal systems.

CHAPTER - VIII

TESTING

8.1 Introduction

Testing is an essential phase in software development to ensure that the developed system functions correctly and meets user requirements. The Stress Detection Web Application was tested at multiple levels to verify functionality, performance, and accuracy.

The testing process includes:

- Unit Testing
- Integration Testing
- System Testing
- Functional Testing

8.2 Unit Testing

Unit testing was performed on individual modules of the system to verify their correctness independently.

Modules Tested:

1. Audio Upload Module
2. Text Input Module
3. Preprocessing Module
4. Feature Extraction Module
5. Machine Learning Prediction Module
6. Dashboard Display Module

Each module was tested with valid and invalid inputs to ensure error handling and correct output generation.

8.3 Integration Testing

Integration testing ensures that all modules work together properly.

Tested Integrations:

- Audio Input → MFCC → Model Prediction

- Text Input → TF-IDF → Model Prediction
- Backend (Flask) → Frontend Dashboard
- Database → History Display

All integrated components communicated successfully using API endpoints.

8.4 System Testing

System testing evaluates the complete system performance in a real environment.

The entire workflow was tested:

User Login
⬇
Input (Audio/Text)
⬇
Prediction
⬇
Result Display
⬇
History Storage

The system performed as expected under normal usage conditions.

8.5 Functional Testing

Functional testing ensures that every feature works according to requirement specification.

Below are sample test cases (Very Important for marks):

Test Case 1: User Login

Test Case ID	**TC_01**
Module	Login
Input	Valid username & password
Expected Output	User logged in successfully
Actual Output	User logged in successfully
Result	Pass

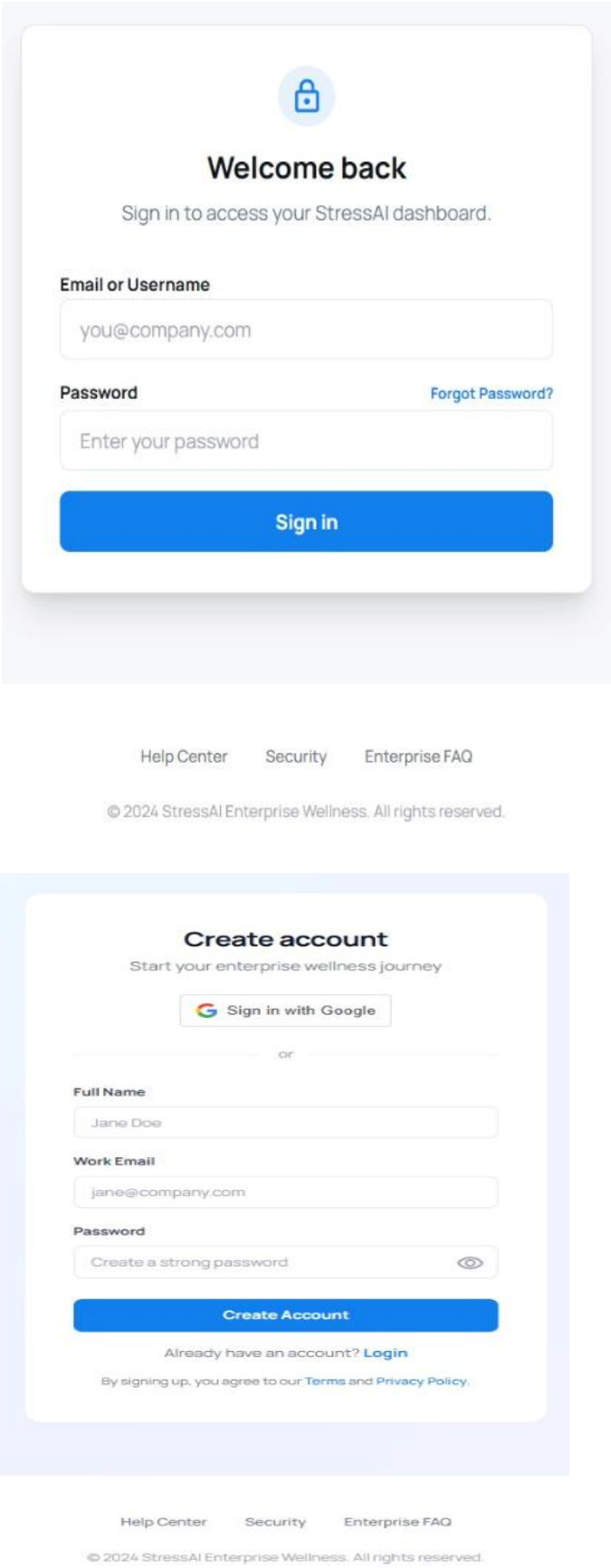

Figure 8.1: User Login Interface and registration

Test Case 2: Audio Upload Prediction

Test Case ID	**TC_02**
Module	Audio Upload
Input	Voice recording
Expected Output	Stress Level Displayed
Actual Output	High Stress
Result	Pass

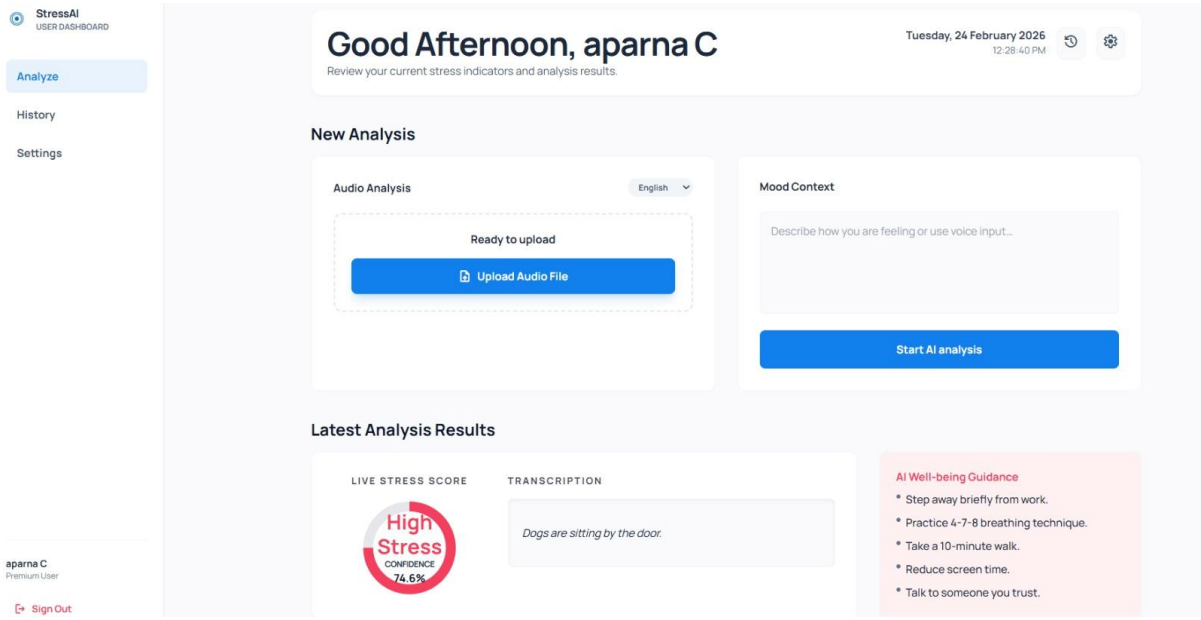

Figure 8.2: Audio-Based Stress Prediction Output

Test Case 3: Text Input Prediction

Test Case ID	**TC_03**
Module	Text Input
Input	“I feel very overwhelmed and tired”

Test Case ID **TC_03**

Expected Output Moderate/High Stress

Actual Output Moderate Stress

Result Pass

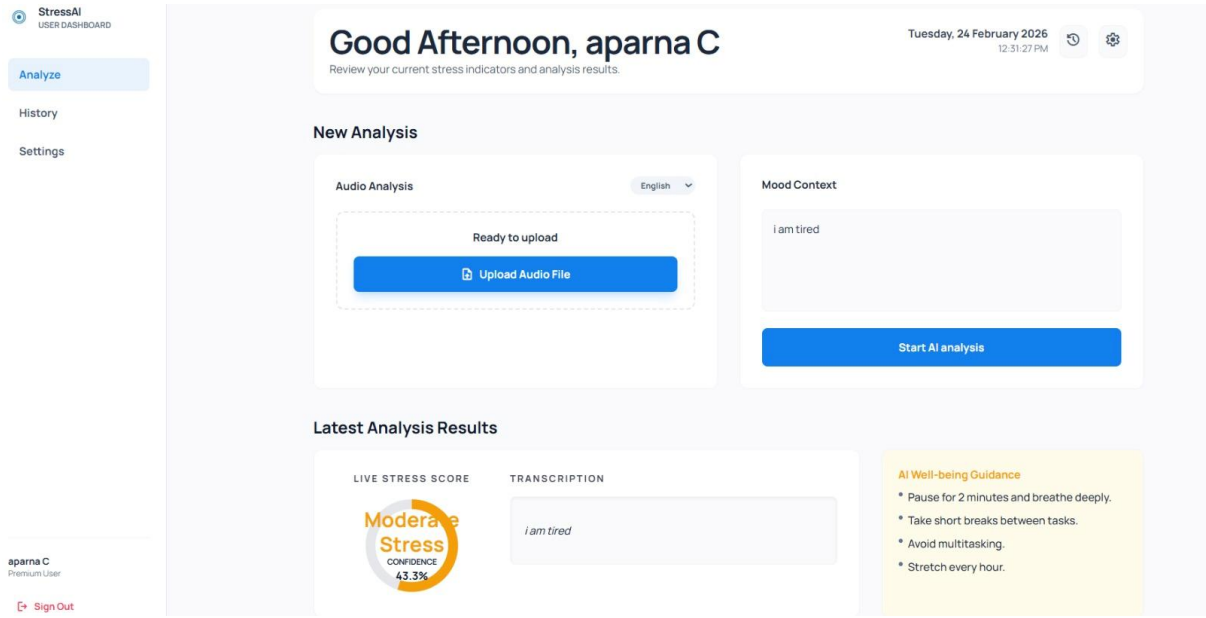

Figure 8.3: Text-Based Stress Prediction Output

Test Case 4: Dashboard Display

Test Case ID **TC_04**

Module Dashboard

Input After prediction

Expected Output Stress Level & Confidence Score

Actual Output Displayed correctly

Result Pass

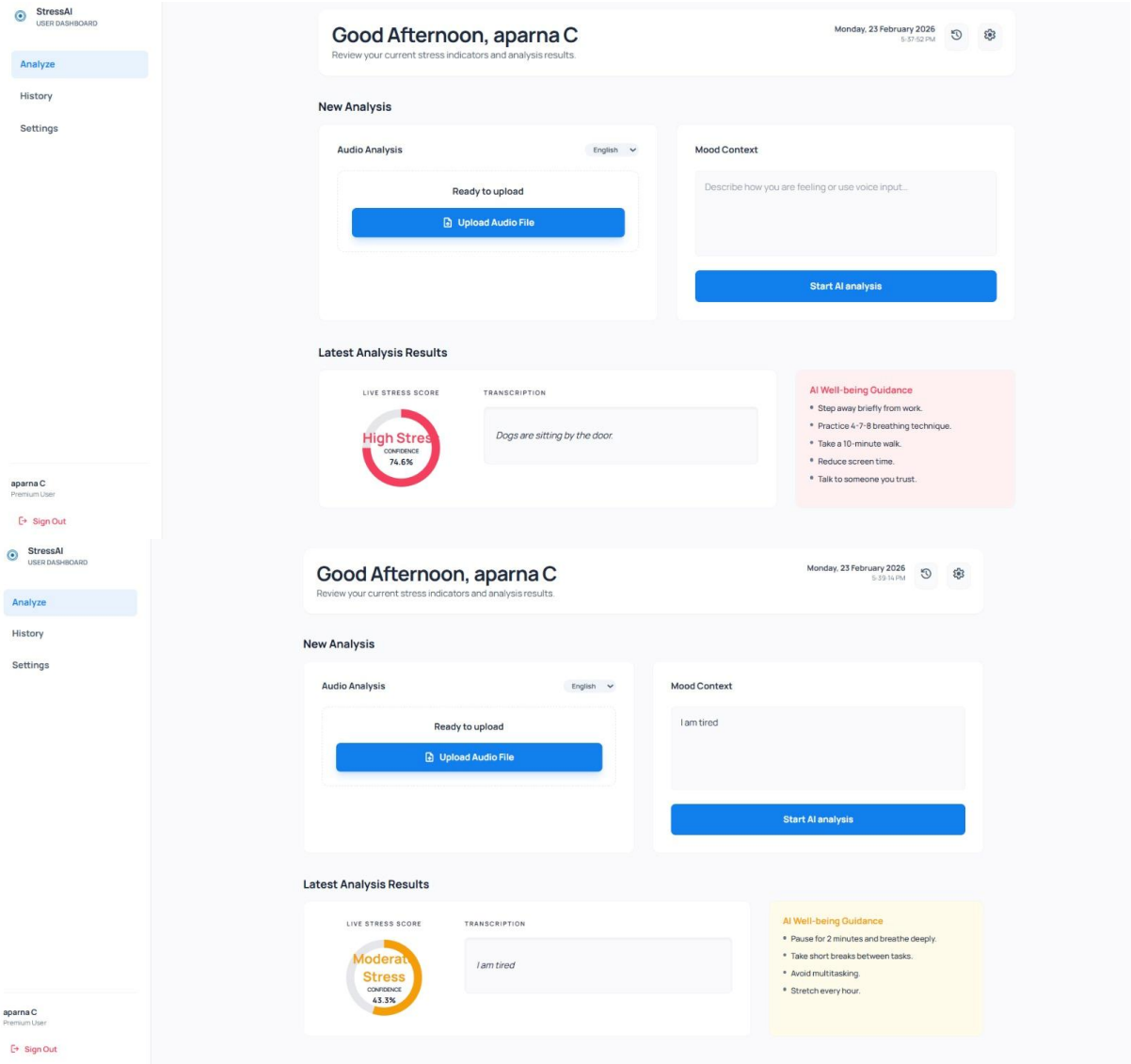

Figure 8.4: Stress Analysis Dashboard

Test Case 5: History Page

Test Case ID	**TC_05**
Module	History
Input	Previous predictions
Expected Output	Stored stress records
Actual Output	Displayed correctly
Result	Pass

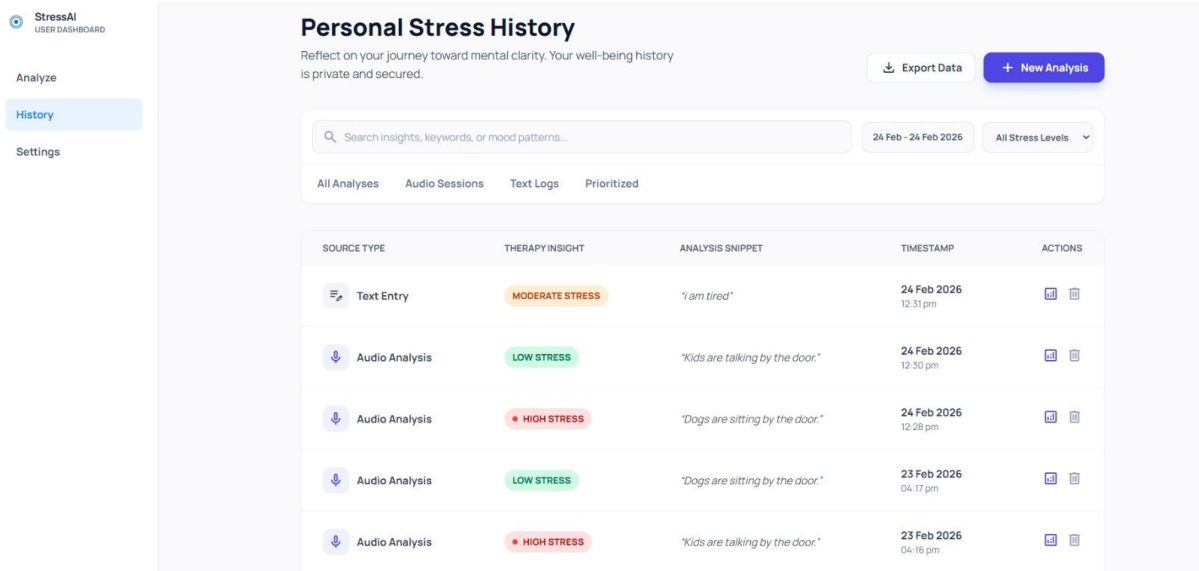

Figure 8.5: User History Page

Test Case 6: Settings Page Verification

Test Case ID TC_06

Module Settings

Input User updates profile / toggles dark mode / changes password

Expected Output Updated preferences saved successfully

Actual Output Changes reflected correctly

Result Pass

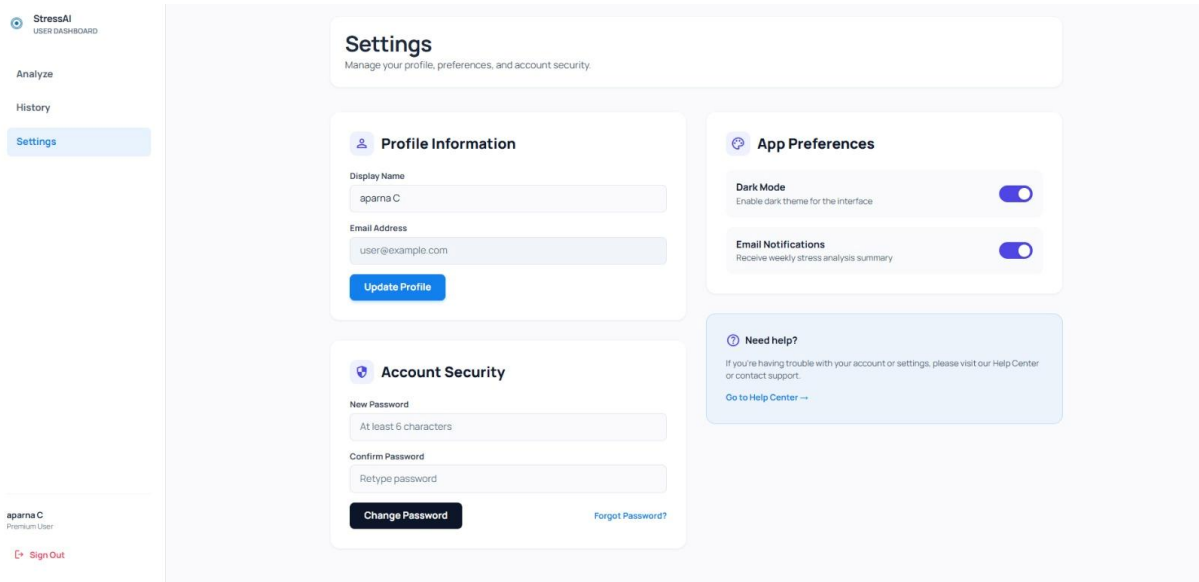

Figure 8.6: Settings Page Interface

8.6 Performance Testing

The system was tested for:

- Response Time
- Processing Speed
- Model Prediction Time

Average prediction time was observed to be within acceptable limits, ensuring real-time user interaction.

8.7 Test Summary

All modules passed functional and integration testing.
No major errors were observed during execution.
The system achieved reliable stress classification and user-friendly interaction.

Test Case 7: Admin Login

Test Case ID	TC_05
Module	History
Input	Previous predictions
Expected Output	Stored stress records
Actual Output	Displayed correctly
Result	Pass

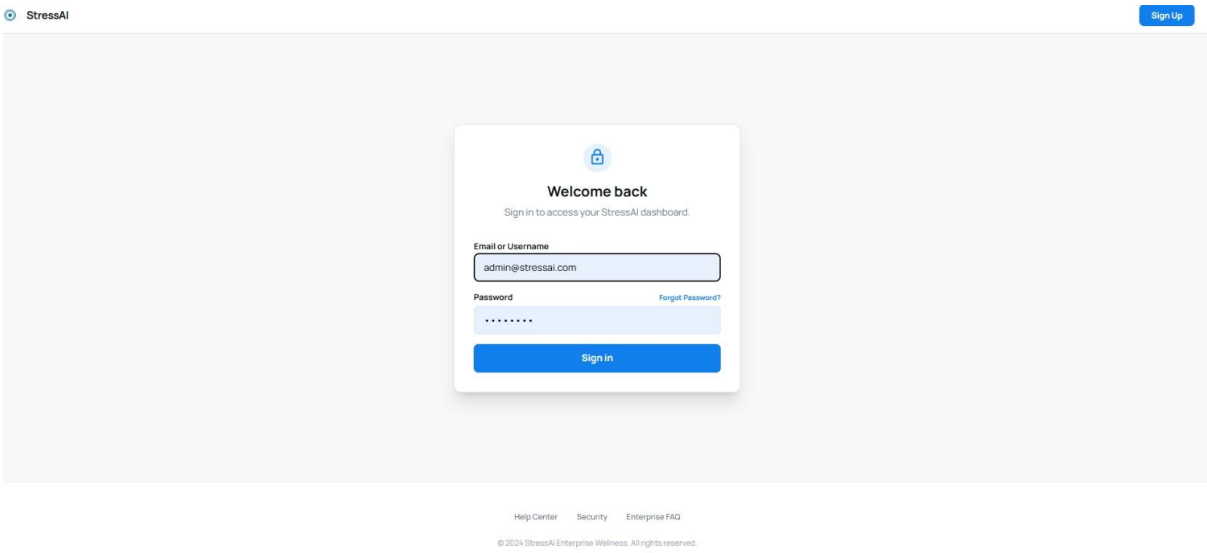

Figure 8.7: Admin Login Interface

Test Case 8: Admin Overview Dashboard

Test Case ID TC_08

Module Admin Overview

Input After admin login

Expected Output Total users, stress rate, usage trends displayed

Actual Output Dashboard loaded correctly

Result Pass

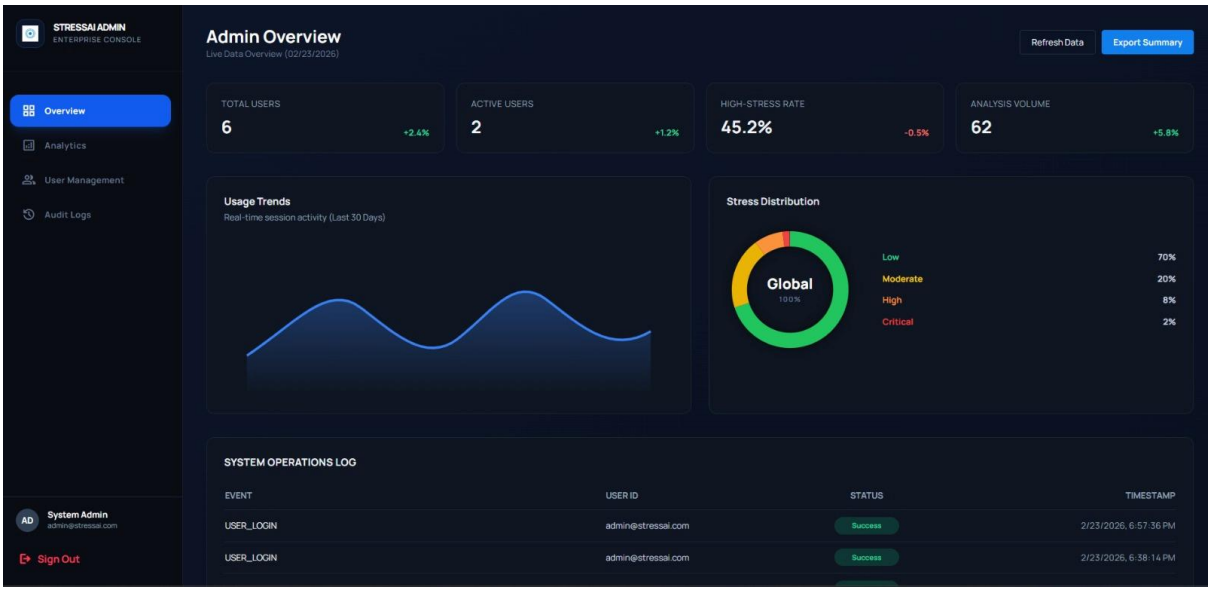

Figure 8.8: Admin Overview Dashboard

Test Case 9: Analytics Module

Test Case ID TC_09

Module Analytics

Input Click Analytics tab

Expected Output Stress distribution charts displayed

Actual Output Analytics report displayed correctly

Result Pass

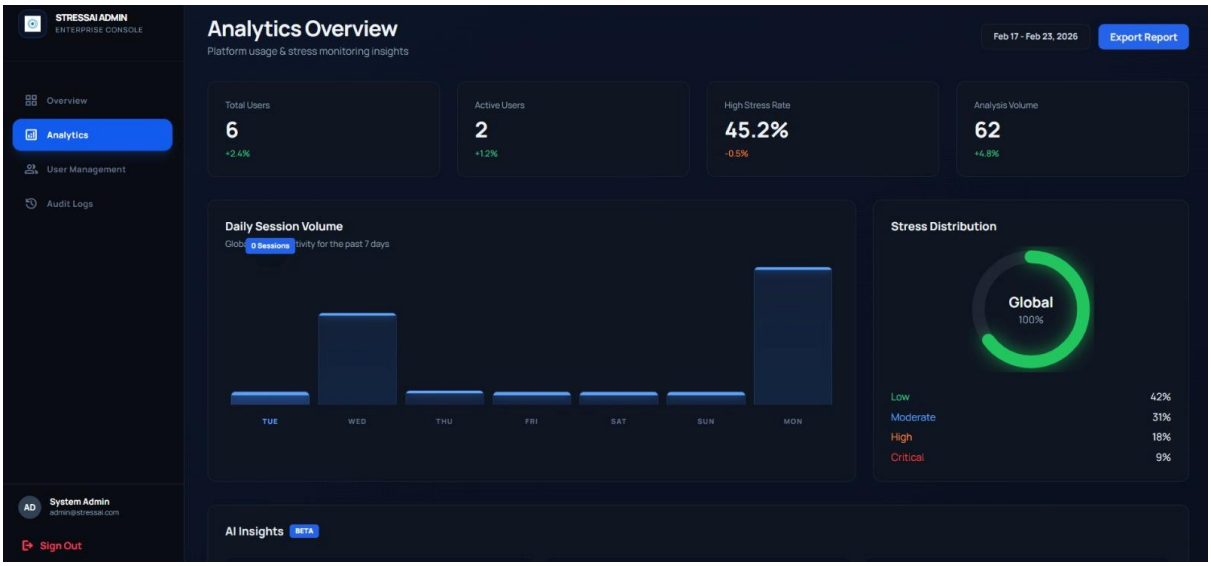

Figure 8.9: Analytics Dashboard

Test Case 10: User Management

Test Case ID TC_10

Module User Management

Input Block / Activate user

Expected Output User status updated

Actual Output Status changed successfully

Result Pass

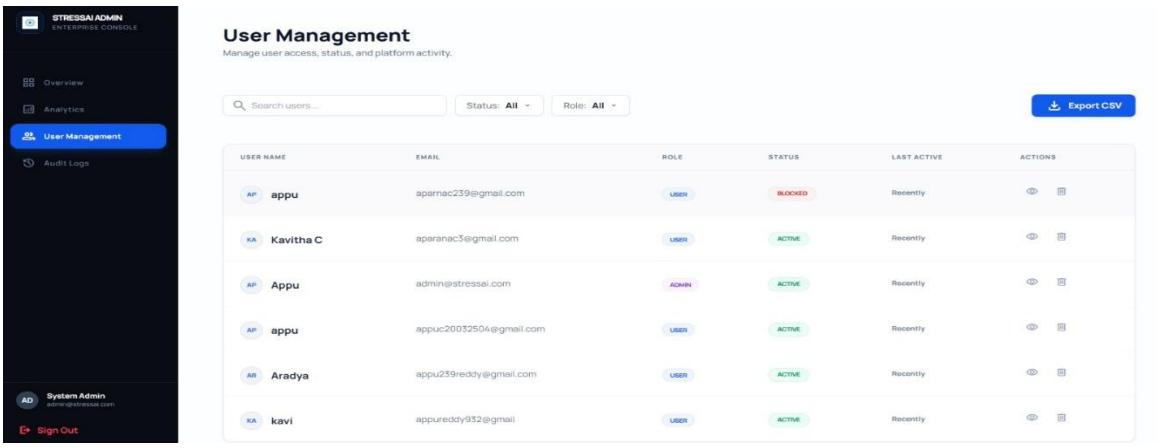

Figure 8.10: User Management Page

Test Case 11: Audit Logs

Test Case ID TC_11

Module Audit Logs

Input View logs

Expected Output System activity logs displayed

Actual Output Logs displayed successfully

Result Pass

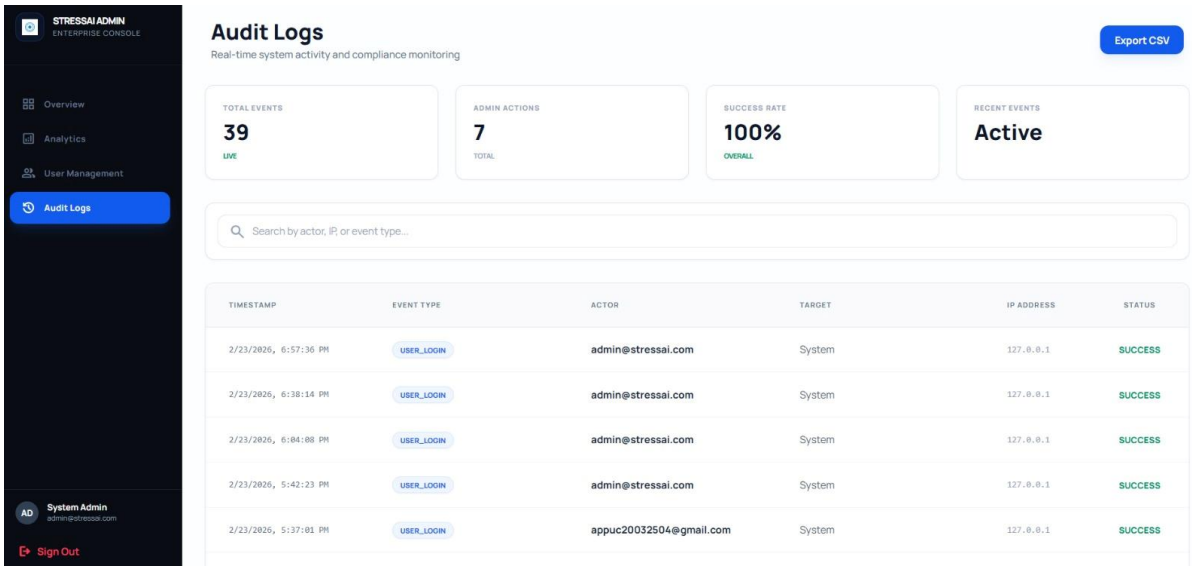

Figure 8.11: Audit Logs Page

CHAPTER -IX

CONCLUSION

The Multimodal AI-Based Stress Detection System developed in this project provides an intelligent and efficient solution for detecting stress levels using both speech and textual inputs. The system integrates audio signal processing and natural language processing techniques to analyze user emotions more accurately compared to traditional single-modal systems.

The application utilizes MFCC (Mel-Frequency Cepstral Coefficients) for extracting speech features and TF-IDF (Term Frequency–Inverse Document Frequency) for text feature representation. These features are then processed using machine learning algorithms such as Random Forest and Support Vector Machine (SVM) to classify stress levels into Low, Moderate, and High categories.

The experimental results demonstrate that the multimodal approach improves prediction reliability and classification performance. The system successfully integrates frontend, backend, and machine learning components into a functional web-based dashboard using Flask and MongoDB.

Testing results confirm that the system performs effectively under different input scenarios and maintains stable performance across multiple user interactions.

Overall, the proposed system provides a scalable, user-friendly, and technically sound solution for AI-based stress detection and can be further enhanced for real-time healthcare and wellness monitoring applications.

FUTURE ENHANCEMENT

The current system can be further improved in several ways:

• The system can be enhanced to support **real-time stress detection** using live microphone input.

• Deep learning techniques such as neural networks can be used to improve prediction accuracy.

• A mobile application can be developed to make the system more accessible.

• Additional datasets can be used to improve model training and performance.

• A graphical user interface can be improved for better user experience.

• Cloud integration can be added to store and analyze stress data.

• Accuracy can be improved by using larger and more diverse datasets

CHAPTER - X

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CHAPTER -XI

APPENDIX

Appendix A – Speech Stress Detection Code (speech_stress.py)

The following code is used to extract speech features and train the stress detection model.

```
import os
import librosa
import numpy as np
import joblib
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

DATASET_PATH = "Stress_Datasets"

LABELS = {
    "neutral": 0,
    "calm": 0,
    "happy": 0,
    "sad": 1,
    "angry": 2,
    "fearful": 2
}

X = []
y = []

def extract_mfcc(file_path):

    audio, sr = librosa.load(file_path, mono=True, duration=3)

    mfcc = librosa.feature.mfcc(y=audio, sr=sr, n_mfcc=40)

    return np.mean(mfcc.T, axis=0)


for root, dirs, files in os.walk(DATASET_PATH):

    for file in files:

        if file.endswith(".wav"):

            file_path = os.path.join(root, file)

            label = None
```

```
        for emotion in LABELS:

            if emotion in file.lower():
                label = LABELS[emotion]
                break

        if label is not None:

            features = extract_mfcc(file_path)

            X.append(features)

            y.append(label)


X = np.array(X)
y = np.array(y)


from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = SVC(kernel="linear", probability=True)

model.fit(X_train, y_train)


joblib.dump(model, "stress_model.pkl")

print("stress_model.pkl created successfully")
```

Appendix B – MFCC Feature Extraction Code

This code is used to extract MFCC features from speech audio files.

```
import librosa
import numpy as np

def extract_mfcc(file_path):

    audio, sample_rate = librosa.load(
        file_path,
        mono=True,
        duration=3
    )

    mfcc = librosa.feature.mfcc(
```

```
        y=audio,
        sr=sample_rate,
        n_mfcc=40
    )

    features = np.mean(mfcc.T, axis=0)

    return features
```

Appendix C – Pseudo Code

START

Load speech dataset

Extract MFCC features from audio files

Assign stress labels

Split dataset into training and testing

Train SVM model

Save trained model

END